

Condensed Mathematics: One Foundation for Algebra and Topology

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Abstract

Condensed mathematics, developed by Clausen and Scholze [1], replaces topological spaces with *condensed sets*: sheaves of sets on the site of profinite sets with finite jointly surjective covers. This small change dissolves the boundary between discrete algebra and continuous analysis. Every topological abelian group becomes a condensed abelian group; the derived category of condensed abelian groups has enough injectives and projectives; the formalism of homological algebra — Ext, Tor, derived functors — extends to all of functional analysis without restriction. The program has two parts. *Condensed mathematics* (Clausen–Scholze 2019–2022) rebuilds commutative algebra and sheaf theory over the condensed site. *Analytic geometry* (Clausen–Scholze 2022 [2]) introduces *liquid* and *solid* modules to handle the analytic and algebraic cases uniformly. The goal is a single language in which perfectoid spaces [4], p -adic Hodge theory, and the global Langlands program [5] can all be stated simultaneously — the universal grammar [6] made concrete.

1 The Problem with Topological Spaces

Classical mathematics runs on two parallel tracks that never fully merge:

- *Algebra*: abelian groups, modules, rings — discrete objects with no topology. Homological algebra (Ext, Tor, derived categories) works perfectly.
- *Analysis*: Banach spaces, Fréchet spaces, locally convex spaces — continuous objects with topology. The naive attempt to do homological algebra fails: Ext^1 between Banach spaces is not well-behaved; the derived category of topological abelian groups has pathological properties.

Proposition 1 (The failure of topological abelian groups). *The category $\mathbf{TA}b$ of topological abelian groups does not have enough projectives. The functor $\mathrm{Hom}_{\mathrm{cts}}(\mathbb{R}, -)$ is not right-exact. The derived functor $\mathrm{Ext}_{\mathrm{cts}}^1(\mathbb{R}, \mathbb{Z})$ is not computable by standard methods. In particular, the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z} \rightarrow 0$ does not split in $\mathbf{TA}b$, but the obstruction cannot be located in a well-defined Ext group.*

Remark 2 (The root cause). The failure traces to the definition of topological space itself. A topological space specifies which subsets are open, but this data does not interact well with algebraic operations: the tensor product of two topological abelian groups is not the coproduct in any standard sense, and limits and colimits behave differently from the algebraic case.

2 Condensed Sets

Definition 3 (Profinite set). A *profinite set* is a compact totally disconnected Hausdorff topological space, equivalently an inverse limit of finite sets: $S = \varprojlim_i S_i$, $|S_i| < \infty$. The category **ProFin** of profinite sets is the opposite of the category of Boolean rings (Stone duality).

Definition 4 (Condensed set [1]). A *condensed set* is a sheaf of sets on **ProFin** with the topology generated by finite jointly surjective families: $\{f_i : S_i \rightarrow S\}$ covers S if $\coprod f_i : \coprod S_i \rightarrow S$ is surjective. Explicitly, a condensed set is a functor $T : \mathbf{ProFin}^{\text{op}} \rightarrow \mathbf{Sets}$ such that $T(\emptyset) = \{*\}$ and for every finite cover $S = S_1 \cup S_2$: $T(S) \xrightarrow{\sim} T(S_1) \times_{T(S_1 \cap S_2)} T(S_2)$.

Proposition 5 (Every topological space gives a condensed set). *There is a fully faithful functor*

$$\iota : \mathbf{Top} \hookrightarrow \mathbf{Cond}(\mathbf{Sets}), \quad X \mapsto \underline{X},$$

where $\underline{X}(S) = \text{Hom}_{\text{cts}}(S, X)$ (continuous maps from the profinite set S to X). For compactly generated Hausdorff spaces, ι is fully faithful and reflects isomorphisms.

Remark 6 (What condensed sets add). A condensed set T knows not just which maps $S \rightarrow X$ exist (topology) but which maps are “continuous in families” parametrized by profinite sets. The profinite sets are the universal compact objects: every compact Hausdorff space is a quotient of a profinite set. Replacing open sets with profinite test objects repairs the failure of topological abelian groups: the sheaf condition is compatible with algebraic operations.

3 Condensed Abelian Groups and Homological Algebra

Definition 7 (Condensed abelian group). A *condensed abelian group* is a sheaf of abelian groups on **ProFin**. The category **Cond(Ab)** is an abelian category with enough projectives and injectives.

Theorem 8 (Clausen–Scholze [1]). *The category **Cond(Ab)** has enough projectives. Specifically, for every profinite set S , the condensed abelian group $\mathbb{Z}[S]$ (the free abelian group on continuous maps from profinite sets to S) is projective. Every condensed abelian group admits a projective resolution by direct sums of such $\mathbb{Z}[S]$.*

Corollary 9 (Homological algebra for analysis). *For condensed abelian groups A, B :*

1. *The derived functors $\text{Ext}_{\text{cond}}^i(A, B)$ are well-defined for all $i \geq 0$.*

2. For $A = \mathbb{R}$, $B = \mathbb{Z}$: $\mathrm{Ext}_{\mathrm{cond}}^1(\mathbb{R}, \mathbb{Z}) \cong \mathbb{R}/\mathbb{Z}$, recovering the correct answer with no pathology.
3. The Künneth formula, the universal coefficients theorem, and the Eilenberg–Steenrod axioms all hold in $\mathbf{Cond}(\mathbf{Ab})$.

4 Solid and Liquid Modules

To handle the analytic case (Banach spaces, p -adic analysis), Clausen and Scholze introduce two full subcategories.

Definition 10 (Solid modules [2]). A condensed abelian group M is *solid* if for every profinite set S , the natural map $M(S) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cond}}(\mathbb{Z}[S], M)$ is an isomorphism. Solid modules are the condensed analogue of complete discrete modules: they see profinite sets as “completed free” objects. The category **Solid** contains all p -adic representations, all discrete abelian groups, and all profinite groups.

Definition 11 (Liquid modules [2]). Fix $0 < r < 1$. A condensed $\mathbb{R}$ -module M is *r -liquid* if for every profinite set S , the natural map $M(S) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cond}}(\mathcal{M}_r(S), M)$ is an isomorphism, where $\mathcal{M}_r(S)$ is the space of r -summable measures on S . The category **Liq_r** contains all Banach spaces over $\mathbb{R}$ and is stable under the full six-functor formalism.

Theorem 12 (Main theorem of analytic geometry [2]). *The derived category $D(\mathbf{Liq}_r)$ provides a simultaneous home for real analysis and algebra:*

1. Every Banach space is liquid. Continuous linear maps are the correct morphisms.
2. The derived tensor product $\otimes^L$ in $D(\mathbf{Liq}_r)$ recovers the completed tensor product of Banach spaces.
3. Sheaf cohomology in $D(\mathbf{Liq}_r)$ over a compact manifold X recovers de Rham cohomology: $R\Gamma(X, \mathbb{R}) \simeq H_{\mathrm{dR}}^*(X)$.

5 Condensed Mathematics and the Global Program

Proposition 13 (Condensed p -adic geometry). *Every perfectoid space [4] is naturally a condensed analytic space in the sense of [2]. The tilting equivalence $(-)^{\flat}$ is a functor of condensed analytic spaces. Prismatic cohomology [4] is a cohomology theory in the condensed framework.*

Remark 14 (Global Langlands in condensed language). Scholze has proposed [3] that the global Langlands correspondence for a reductive group G over a number field F should be stated as an equivalence in the derived category of condensed modules on the stack of global L -parameters $\mathrm{Loc}_{\hat{G}}(\mathrm{Spec} \mathbb{Z})$. The condensed framework provides the correct topology on $\mathrm{Spec} \mathbb{Z}$ (via its profinite covers) to make this stack well-defined. This is the global analogue of the Fargues–Scholze theorem [4]: local Langlands via the Fargues–Fontaine curve, global Langlands via a condensed stack over $\mathrm{Spec} \mathbb{Z}$. The global program remains conjectural; condensed mathematics is the language in which it is supposed to be stated.

Theorem 15 (Position in the hierarchy).

<i>Framework</i>	<i>What it provides</i>
<i>Topological spaces</i>	<i>Continuous geometry (incomplete for homological algebra)</i>
<i>Condensed sets</i>	<i>Unified foundation for algebra and analysis</i>
<i>Solid modules</i>	<i>p-adic and discrete algebra</i>
<i>Liquid modules</i>	<i>Real and complex analysis</i>
<i>Condensed analytic geometry</i>	<i>Perfectoid spaces, prismatic cohomology</i>
<i>Global condensed Langlands</i>	<i>Conjectural:</i> <i>language for global Langlands</i>

Each row is supposed to be alive — the bottom row is the one still becoming so.

References

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